INFO 4220 – Networks II: Market Design Spring 2025 – Problem Set 4

Solution

Problem 1 (20 points)

The group of friends from PS3—**Alex, Jordan, Sam, and Riley**— did not like how the first D&D campaign went with the classes they got using probabilistic assignment. The group will start a second campaign without Riley, who is out of town. They decided they would all be multi-class for a second campaign, and no class should be used more than twice. The preferences are the following:

Alex	**Jordan**	**Sam**
Fighter	Fighter	Rouge
Rouge	Rouge	Wizard
Wizard	Wizard	Fighter
Cleric	Cleric	Cleric

a. Alex is taking INFO4220, so he proposes using the HBS draft mechanism. Find all the possible class assignments and the probability that each occurs assuming that students report their truthful preferences. (4 pts)

ANSWER: There are 6 possible orders for Alex, Jordan, and Sam. We can determine the assignment for each:

Order	Alex	Jordan	Sam
A, J, S	Fighter, Wizard	Fighter, Rouge	Rouge, Wizard
A, S, J	Fighter, Wizard	Fighter, Rouge	Rouge, Wizard
J, A, S	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
J, S, A	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
S, A, J	Fighter, Wizard	Fighter, Rouge	Rouge, Wizard
S, J, A	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard

2 points

The assignment Alex – {F, W}, Jordan – {F, R}, and Sam – {R, W} has probability ½ **1 point**

And Alex – {F,R}, Jordan – {F, W}, and Sam – {R, W} has probability ½ **1 point**

b. Now assume that Alex, knowing the mechanism is not strategyproof, decides to misrepresent his preferences (assume Jordan and Sam report truthful preferences). Find Alex's optimal strategy (and explain why he should use it) and find all the possible class assignments and the probability that each occurs. (4 pts)

ANSWER: Alex is not getting Wizard instead of Rouge half of the times because in those cases Jordan is getting it. He can realize that rather than listing Fighter first, he can list Rouge first. If he does this, what will happen is:

Order	Alex	Jordan	Sam
A, J, S	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
A, S, J	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
J, A, S	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
J, S, A	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
S, A, J	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
S, J, A	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard

So now Alex always gets Fighter, Rouge, and Jordan always gets Fighter, Wizard.

2 points for explaining the strategy
2 point for the assignments

c. If Alex reports his preferences as you determined it was optimal in part (b), should Jordan report his true preferences, or should he also be strategic and misrepresent his preferences? If so, what should he do and what all the possible assignments that could result along with its probabilities. (3 pts)

ANSWER: Jordan now is never getting his 2 most preferred courses, so he can try to do the same strategy as Alex and put Rouge first. If he does that:

Order	Alex	Jordan	Sam
A, J, S	Rouge, Wizard	Rouge, Fighter	Wizard, Fighter
A, S, J	Rouge, Fighter	Fighter, Wizard	Rouge, Wizard
J, A, S	Rouge, Fighter	Rouge, Wizard	Wizard, Fighter
J, S, A	Fighter, Wizard	Rouge, Fighter	Rouge, Wizard
S, A, J	Rouge, Fighter	Fighter, Wizard	Rouge, Wizard
S, J, A	Fighter, Wizard	Rouge, Fighter	Rouge, Wizard

He is better with respect to part (b) as now he is getting Fighter, Rouge 4 times.

1 points for the strategy
2 points for the assignments

d. Based on your answers to parts (b) and (c), do you expect Alex and Jordan to misrepresent their preferences. (3 pts)

ANSWER: Alex will misrepresent his preferences.
1 point

For Jordan is not as obvious:
We know that in (b) Jordan gets the first and third class always.

In (c) he will get:

In 3 cases something that he likes better (Rouge, Fighter)

And in 1 case something that he likes worse (Rouge, Wizard)

Depending on how much worse he likes (Rouge, Wizard) and his risk aversion, he may choose to misrepresent their preferences or not. If we use an assumption like in part (f) we can conclude he would.

If we go strictly by responsive preferences, then they wouldn't. (Both answers are considered correct as they are both logical)

2 points

e. Are the schedules you found in part (c) efficient? Why or why not? (3 pts)
No, they are not. For example, for the result of the order A, S, J, Alex and Sam could exchange a class to be better off. Alex can give Sam the Rouge in exchange of the Fighter.
1 point not efficient
2 points for explanation

f. Jordan proposes using random serial dictatorship (with students choosing their full schedules on their turn) instead of HBS since it is strategyproof. Find all assignments that could result and the probability of each. Compare what you obtain with the results of part (a). You can assume that a student gets a utility of 4 for their most preferred course, 3 for their second most preferred course, 2 for their third most preferred course, and 1 for their least preferred course (4 pts)

Order	Alex	Jordan	Sam
A, J, S	Fighter, Rouge	Fighter, Rouge	Wizard, Cleric
A, S, J	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
J, A, S	Fighter, Rouge	Fighter, Rouge	Wizard, Cleric
J, S, A	Fighter, Wizard	Fighter, Rouge	Rouge, Wizard
S, A, J	Fighter, Rouge	Fighter, Wizard	Rouge, Wizard
S, J, A	Fighter, Wizard	Fighter, Rouge	Rouge, Wizard

With the utilities we can calculate the expected utility. With true preferences Alex and Jordan get a expected utility of 6.5, and with the RSD 6.67. So, they are slightly better off. Sam instead is much worse. With true preferences Sam gets a expected utility of 7, and with RSD 6.
1 point for assignments
1 point for each expected utility (3 points in total)

(Note: As the problem did not clearly state that students could take up 2 courses, if anyone answered the question using no limit on the number of courses, those answers were graded on a case-by-case basis consistent with that assumption)

Problem 2 (10 points, 2.5 pts each)

Four new Pokémon trainers need to choose their starter Pokémon. Professor Oak uses the Gale-Shapley Pareto Dominant Market Mechanism to allocate the Pokémon. The preferences students submitted are as follows:

Ash	**Misty**	**Brooks**	**Gary**
Pikachu	Squirtle	Bulbasaur	Pikachu
Charmander	Bulbasaur	Charmander	Squirtle
Squirtle	Charmander	Squirtle	Bulbasaur

The student submits the following bids:

Bid	Pikachu	Charmander	Squirtle	Bulbasaur
Ash	1	100	0	0
Misty	0	0	100	1
Brooks	0	50	0	51
Gary	1	0	100	0

To break the ties, Professor Oak decides to do it alphabetically, with Ash, Brooks, Gary, and Misty as the priority.

a. Determine the Pokémon allocation in this scenario, explaining and showing each step of the algorithm.

ANSWER:

Step 1:
Ash – Pikachu
Misty – Squirtle
Brooks – Bulbasaur
Gary – Pikachu

Pikachu has 2 offers. Ash and Gary had both bid the same, so we reject Gary based on the rule of rejecting by alphabetic order.

2 point

Step 2:
Gary – Squirtle

Now Squirtle has 2 offers. Misty and Gary had bid the same, so we reject Misty based on the alphabetic order rule

1 point

Step 3:
Misty – Bulbasaur. No one is rejected.

Assignment:
Ash – Pikachu
Misty – Charmander
Brooks – Bulbasaur
Gary – Squirtle

1 point

Misty believes that Oak's tie breaking mechanism is unfair. She proposes that it should be a lottery instead with the following result (lowest number wins when breaking ties):

Ash	Misty	Brooks	Gary
2	3	1	4

b. Does the allocation change?

ANSWER:

Step 1:
Ash – Pikachu
Misty – Squirtle
Brooks – Bulbasaur
Gary – Pikachu

Pikachu has 2 offers. Ash and Gary had both bid the same, so we reject Gary based on the lottery rule.
1 point

Step 2:
Gary – Squirtle

Now Squirtle has 2 offers. Gary gets rejected because of the lottery rule.

1 point

Step 3:
Gary – Bulbasaur.

Gary gets rejected because of the lottery rule.

Assignment:
Ash – Pikachu
Misty – Squirtle
Brooks – Bulbasaur
Gary – None

1 point

d. Can anyone improve their assignment by misrepresenting their preferences? Are there other strategies that they could use to improve their assignment?

Misrepresenting preferences is not a good idea, as it will not improve the allocation. What could work is changing the bids. **1.5 preferences 1.5 bids**

Problem 3 (20 points)

Consider a school district with four schools, each having a capacity of 1 student. Four students—Alice, Bob, Carol, Denis —apply to these schools. Their preferences over the schools, as well as the schools' weak priority orders over the students, are given below.

Alice	Bob	Carol	Denis
S_2	S_2	S_2	S_4
S_3	S_1	S_4	S_2
S_4	S_4	S_3	S_3
S_1	S_3	S_1	S_1

S_1	S_2	S_3	S_4
[Alice, Bob]	Denis	Carol	[Alice, Carol]
Carol	[Alice, Bob, Carol]	Alice	Bob
Denis		Denis	Denis
		Bob	

Please remember to show and explain all your steps in all parts, and clearly show the result. For parts (a) – (d) of this problem assume schools break the ties using a single tie-breaking rule, and that the lottery resulted in the order Alice > Bob > Carol > Denis.

a. Consider the assignment μ': '(Alice) = S_3, μ'(Bob) = S_1, μ'(Carol) = S_4, μ'(Denis) = S_2. Is the assignment stable? Explain why or why not (3 pts)

ANSWER: The assignment is stable. Because it is individual rational, non-wasteful and no justified envy.

1 point for saying it is stable, 2 for explaining.

b. Use DA with the student proposing to find the assignment. (3 pts)

ANSWER:

Step 1: Alice -> S2, Bob -> S2, Carol -> S2, Denis -> S4. Bob and Carol are rejected by S2.
Step 2: Bob -> S1, Carol -> S4. Denis is rejected by S4 because S4 prefers Carol than Denis.
Step 3: Denis -> S2. Alice is rejected by S2.
Step 4: Alice -> S3. No more rejection.
Final assignment: Alice -> S3, Bob -> S1, Carol -> S4, Denis -> S2.
1 point for the assignment, 2 points for the step by step.

c. Is the assignment you found in (b) efficient? Explain why. (3 pts)

ANSWER: It is not efficient. Because Carol and Denis can trade to make both of them better.

1 point for saying it is not efficient, 2 points for explaining.

d. Now try to improve the assignment you found in (b) using improvement cycles. (3 pts)
Trade cycle between Carol and Denis.

ANSWER: Final assignment: Alice -> S3, Bob -> S1, Carol -> S2, Denis -> S4.
1 point for the assignment, 2 points for explaining.

e. If students know that DA + improvement cycles will be used, is there a way Alice could misrepresent her preferences to obtain a better assignment? If so, show Alice's optimal strategy and explain why. (4 pts)

ANSWER: Alice could misrepresent her preference as: S2, S4, S3, S1. **(1 point)** In this way, it is possible that she could be assigned to S2 after trading with Denis, because she has the same priority in S2 and S4 as Carol. **(2 points)** If not misrepresenting her preference and everyone submits their true preferences, regardless the order from tie breaking, it will always be Carol that assigned to S4 after DA and be able to trade with Denis. **(1 points)** (Might need to discuss more comprehensive scenario for full credits.)

The following two sub questions are independent from the questions above. Answer them based on what we saw/discussed in class:

f. We need to decide whether to choose single tie-breaking (STB) or multiple tie-breaking (MTB) for the assignment problem. Simon says, "multiple tie-breaking (MTB) is better because it lets each school resolve ties independently, offering varied outcomes that can favor different students." Do you agree with this? What method would you recommend to use? Explain your choice by discussing the trade-off among fairness, optimality, ease of implementation, etc. (2 pts)

ANSWER: I disagree with Simon. **(1 point)** MTB seems fairer but would not generate the student-optimal results. We will not lose much by using STB as the assignments could only generate by MTB would not be the "good" assignment. **(1 point)**

g. A policymaker is considering limiting the number of schools that each student can submit on their application. Identify 1 potential benefit and 1 potential downside of imposing such a constraint. Provide a brief explanation for each. (2 pts)

ANSWER: A potential benefit could be that it is easier to implement the algorithm as it makes it less complex. And it makes it "easier" for students/parents to decide on the list to submit because it is shorter (though it might not be their best preference). **(1 point)**

The downside would be that it might lead students/parents to be strategic, as they might start considering schools they would otherwise not consider, which could potentially harm the fairness. **(1 point)**